Supplementary Information for Surana et. al, PGViS

Supplementary Figures

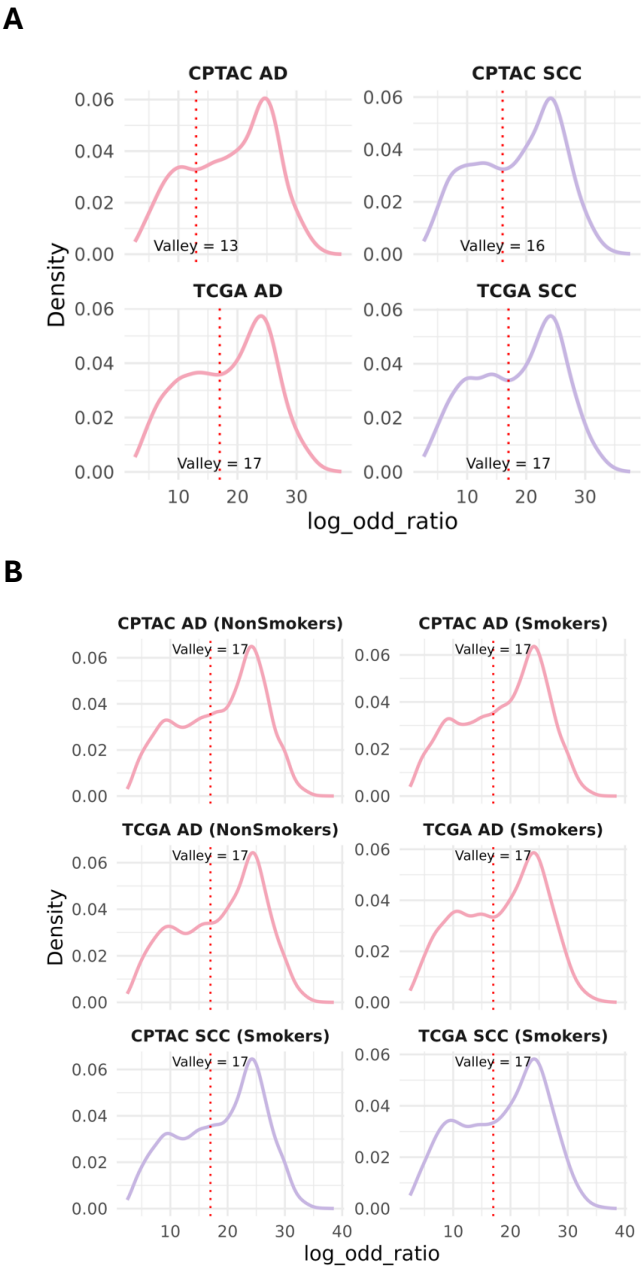


Fig. S1: Density plot of the DNABERT-derived $log_2\ OR$ for all analyzed variants. The distribution shows an approximately bimodal pattern, indicating two distinct groups of variant effects. An initial DNABERT LOR cutoff was applied to identify potentially significant mutations, followed by additional thresholding to refine the final mutation set. The right-hand peak of the distribution, corresponding to variants with higher predicted regulatory impact, was used as a key threshold for selecting significant mutations included in downstream analyses.

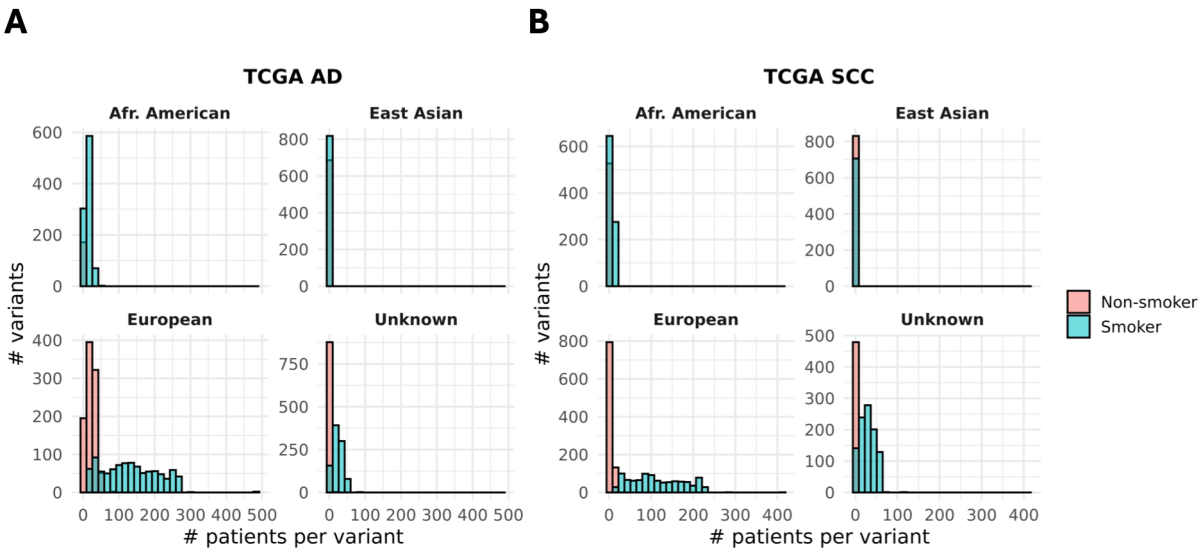


Fig. S2: Distribution of variant burden per patient across ancestry groups in TCGA cohort (A) AD and (B) SCC stratified by smoking status shows clear distinction between smokers and nonsmokers across ancestries.

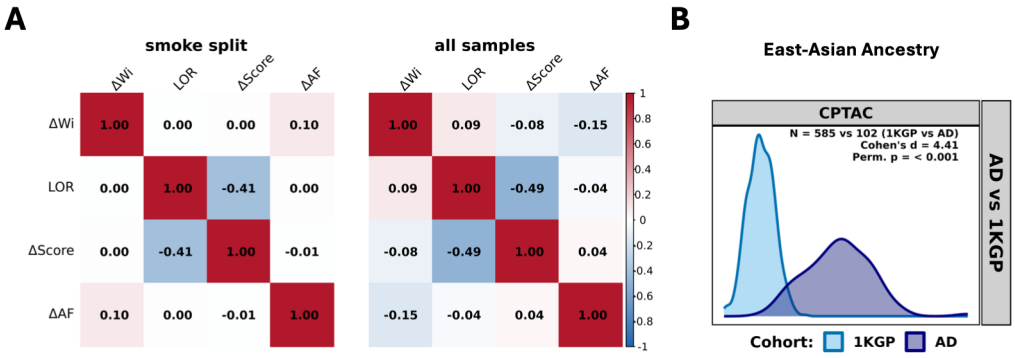


Fig. S3: PGViS score distributions for EAS-ancestry samples. CPTAC AD patients (n = 102) versus the 1kGP EAS reference (n = 585). Cohen's d, and permutation p-value are annotated. TCGA and SCC comparisons are not shown for EAS owing to insufficient sample size. *Note*: ΔScore is alternatively used as ΔAF_{alt}

Supplementary Tables

Table S1. Number of samples in AD and SCC of lung across CPTAC and TCGA cohorts. The % is with respect to the total samples in each cohort.

WGS dataset	Lung AD				Lung SCC				Total Across
	Smoker	NoSmoker	Unknown	Total	Smoker	NoSmoker	Unknown	Total	
CPTAC	133	87	8	228 (68%)	84	16	9	109 (32%)	337
TCGA	362	61	13	436 (57%)	304	14	11	329 (43%)	765
Total	495	148	21	664	388	30	20	438	1102

Table S2: Table summarizing those variants where the ref and alt alleles as the positions match between cohort (CPTAC, TCGA) and dbSNP v155. This is the starting group of variants across functional sites used in this study.

Cancer:	Cohort	Number of sites across TFBS + Splice Region		
		Unique Functional Site	# Variants	dbSNP IDs
ADD	TCGA: all variants	16431811	8080612	8034254
	CPTAC	15555095	6057649	6031724
	TCGA ∩ CPTAC	13081924	3980558	3969128
SCC	TCGA: all variants	15971353	6949822	6914833
	CPTAC	13850252	4383867	4369786
	TCGA ∩ CPTAC	11974579	3334336	3326272

Table S3: Summary of thresholds used to arrive at final list of significant mutations used further in the analysis. This analysis is done combining all samples in the cohort without classification further based on clinical features. (A) Thresholds used to get significant variants across functional sites for both AD and SCC in the CPTAC cohort. (B) Thresholds used to get significant variants across functional sites for both AD and SCC NSCLCs in the TCGA cohort.

A

Cancer	Thresholds	Significant # of across TFBS + Splice Region		
		Unique Functional Site	CPTAC Variants	dbSNP IDs
AD	Total Sites	15866421	7132420	7589626
	Total short variants (<10 bp)	15555095	6057649	6031724
	Variants in >= 5% of patients	8517238	2002981	2000316
	abs(ΔAF_{alt}) >= 0.1	2131794	387076	386650
	ΔWi_{INT} >= 0	179084	33469	33429
	ref_prob > 0.5 & alt_prob <= 0.5	9483	5392	5389
	ref_prob > 0.7 & alt_prob <= 0.3 And Exclude M/mAF<0.05	3692	2161	2159
	LOD >= 13 (Valley inflection)	2783	1721	1720
SCC	Total Sites	14294688	5214003	5686700
	Total short variants (<10 bp)	13850252	4383867	4369786
	Variants in >= 5% of patients	8571198	2020941	2018204
	abs(ΔAF_{alt}) >= 0.1	1209636	220587	220154
	ΔWi_{INT} >= 0	139193	31723	31530
	ref_prob > 0.5 & alt_prob <= 0.5	8185	4943	4930
	ref_prob > 0.7 & alt_prob <= 0.3 And Exclude M/mAF<0.05	3709	2295	2288
	LOD >= 16 (Valley inflection)	2304	1578	1572

B

Cancer:	Thresholds	Significant Number of across TFBS + Splice Region		
		Unique Functional Site	**TCGA Variants**	**dbSNP IDs**
AD	Total Sites	16599737	9500957	9876505
	Total short variants (<10 bp)	16431811	8080612	8034254
	Variants in >= 5% of patients	8459283	2001368	1998783
	abs(ΔAF_{alt}) >= 0.1	827197	152884	152443
	ΔWi_{INT} >= 0	81830	22796	22541
	ref_prob > 0.5 & alt_prob <= 0.5	5156	3354	3335
	ref_prob > 0.7 & alt_prob <= 0.3 And Exclude M/mAF<0.05 *(Common variants kept)*	2327	1506	1490
	LOD >= 17 (Valley inflection)	1363	965	955
	Total Sites	16200808	8628213	8833051
SCC	Total short variants (<10 bp)	15971353	6949822	6914833
	Variants in >= 5% patients & ref and alt match b/w dbSNP, cohort	8515880	2014785	2012017
	abs(ΔAF_{alt}) >= 0.1	1084527	199095	198651
	ΔWi_{INT} >= 0	90377	24772	24550
	ref_prob > 0.5 & alt_prob <= 0.5	5650	3689	3673
	ref_prob > 0.7 & alt_prob <= 0.3 And Exclude M/mAF<0.05	2353	1524	1511
	LOD >= 17 (Valley inflection)	1391	995	985

Table S4: The table summarizes overlapping variants identified between cohorts, highlighting shared mutational features that may contribute to common biological mechanisms/ risk across patient groups.

	Cancer:	Cohort-wise significant variants	Significant Number of across TFBS + Splice Region		
			Unique Functional Site	# Variants	dbSNP IDs
All samples analyzed together	**AD**	TCGA	1363	965	955
		CPTAC	2783	1721	1720
		TCGA ∩ CPTAC	334	246	245
	SCC	TCGA	1391	995	985
		CPTAC	2304	1578	1572
		TCGA ∩ CPTAC	826	604	600
Smoking history-based separation	**AD Smoker**	TCGA	2309	1598	1593
		CPTAC	8352	5483	5481
		TCGA ∩ CPTAC	820	590	588
	SCC Smoker	TCGA	2365	1625	1620
		CPTAC	8766	5703	5704
		TCGA ∩ CPTAC	982	680	681
	AD NonSmoker	TCGA	4830	3092	3096
		CPTAC	18266	11789	11788
		TCGA ∩ CPTAC	765	512	512

Table S5: Summary of genetic variants used as input features for the PGViS score. Variants were selected and incorporated into the patient-level bag of mutations according to analysis context: smoker-specific, non-smoker-specific, and pooled cohorts. *"All" denotes analysis without dividing samples by clinical variables.

Dataset	Cancer (#Patients)	Significant Number of across TFBS + Splice Region		
		Unique Functional Site	Variants	dbSNP IDs
TCGA	AD Smoker (362)	2309	1598	1593
	AD Non-Smoker (61)	4830	3092	3096
	SCC Smoker (301)	2365	1625	1620
CPTAC	AD Smoker (133)	8352	5483	5481
	AD Non-Smoker (87)	18266	11789	11788
	SCC Smoker (84)	8766	5703	5704
CPTAC	All AD (227)	2783	1721	1720
	All SCC (109)	2304	1578	1572

TCGA	All AD (434)	1363	965	955
	All SCC (329)	1391	995	985
Note: *All is analysis without dividing samples based on clinical variables.				

Table S6: Mapping for significant mutations common to both TCGA & CPTAC cohorts for the European ancestry. These were used to find smoking-dependent & -independent scores which was combined to generate PGViS for every patient. rGREAT: used to map the functional sites to nearest genes.

Dataset (EUR)	Cancer	Type (# patients)	Germline mutations	dbSNP ID	Functional sites	Unique TF name	Unique Genes
CPTAC	**AD**	**NonSmoker (12)**	480	480	708	228	320
		Smoker (74)	487	487	658	220	225
		All* (89)	181	180	233	132	98
	SCC	**Smoker (63)**	567	567	792	231	313
		All* (70)	449	449	590	219	224
TCGA	**AD**	**NonSmoker (46)**	480	480	708	228	320
		Smoker (272)	487	487	658	220	225
		All* (328)	181	180	233	131	98
	SCC	**Smoker (222)**	567	567	792	231	313
		All* (240)	449	449	590	218	224

Note:

1. Some of the genes may not map to any sites within the mutations so the number might drop from the initial tables.
2. *All is analysis without dividing samples based on clinical variables.